

TAD_FINAL_PROJECT_By_Mercy_Ogbede

Patient Experiences and Outcomes After Myomectomy: A Text Analysis of Online Narratives

Introduction

Uterine fibroids, also known as leiomyomas, are non-cancerous growths of the uterus that affect up to 70% of women of reproductive age (Stewart, 2015). For many, the condition causes heavy bleeding, severe menstrual pain, abdominal swelling, and fertility challenges. A myomectomy, which involves surgically removing fibroids while preserving the uterus, is often chosen by women who wish to relieve symptoms and maintain reproductive potential. Despite being a well-established medical procedure, there is limited research capturing how women personally describe their experiences of recovery, pain, and emotional adjustment following surgery.

In recent years, online communities have become essential spaces where individuals share first-hand stories about illness and healing. Platforms such as Reddit allow women to discuss sensitive health issues anonymously, providing insights into lived experiences that traditional clinical studies may overlook (De Choudhury et al., 2014). The subreddit r/Fibroids serves as a virtual support group where patients exchange information, seek reassurance, and reflect on both positive and negative aspects of myomectomy recovery. These discussions often extend beyond physical outcomes to include emotional well-being, body physical condition, and fertility concerns.

This study examines how women narrate their recovery journeys after myomectomy through Reddit posts. By analyzing language patterns and common themes such as pain, bleeding, fertility, and recovery, the research aims to uncover both the physical and emotional dimensions of healing. Using basic Natural Language Processing (NLP) techniques, including text cleaning, tokenization, and word-frequency analysis, this project explores how online patient narratives reflect shared challenges, resilience, and hope. The goal is to complement existing medical literature with patient-driven perspectives that highlight what recovery truly feels like from the women who experience it.

Research Question

This study addresses the following research question:

How do women describe their experiences of recovery, symptoms, and emotional well-being following myomectomy in on Reddit, and what themes dominate these patient-authored discussions

Hypothesis

Before analyzing the data, I hypothesized that:

Most women would describe improvement after myomectomy such as reduced pain or bleeding.

Data Retrieval and Structure

Here, my goal was to collect authentic patient authored narratives about myomectomy recovery. I retrieved recent posts from r/Fibroids using Reddit's public API. After extracting titles, body texts, and metadata, I saved the dataset as a CSV file for reproducibility and reloaded it into R for analysis. This ensures that all subsequent analysis uses a stable dataset rather than re-scraping new posts each time. The final dataset contained 25 individual posts, each representing one patient's narrative. Some were brief, while others contained long descriptions of symptoms, emotions, and follow-up concerns. This dataset became the foundation for the entire analysis, each post treated as a separate "document" similar to a chapter in a book.

Text Cleaning and Corpus Construction

Before analyzing text, it must be cleaned. Reddit posts contain URLs, emojis, punctuation, and inconsistent encoding. Cleaning ensures that the analysis reflects meaningful language rather than noise.

```
#Loading packages
library(jsonlite)
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.3.2

Warning: package 'tidyr' was built under R version 4.3.2

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.1.4      ✓ readr      2.1.5
✓ forcats    1.0.0      ✓ stringr    1.5.1
✓ ggplot2    3.5.1      ✓ tibble     3.2.1
✓ lubridate  1.9.3      ✓ tidyr      1.3.1
✓ purrr      1.0.2
— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ purrr::flatten() masks jsonlite::flatten()
✖ dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(quanteda)
```

Warning: package 'quanteda' was built under R version 4.3.3

Package version: 4.3.1
 Unicode version: 14.0
 ICU version: 71.1
 Parallel computing: disabled
 See <https://quanteda.io> for tutorials and examples.

```
library(quanteda.textplots)
```

Warning: package 'quanteda.textplots' was built under R version 4.3.3

```
library(SnowballC)
# ---- LOAD SAVED REDDIT DATA ----
posts_df <- read_csv("reddit_fibroids_raw.csv")
```

Rows: 25 Columns: 4

— Column specification —————

Delimiter: ","

chr (4): title, text, subreddit, full_text

i Use `spec()` to retrieve the full column specification for this data.
 i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
head(posts_df)
```

A tibble: 6 × 4

title	text	subreddit	full_text
<chr>	<chr>	<chr>	<chr>
1 Need help	"Is this...	Fibroids	"Need he...
2 Slynd. TXA. Aeygestin	<NA>	Fibroids	"Slynd. ...
3 Pain management post op	"Hi! I f...	Fibroids	"Pain ma...
4 Need some hope	"Hi Ever...	Fibroids	"Need so...
5 Pregnancy stories/experiences with large fibroid	"Hi ladi...	Fibroids	"Pregnan...
6 Massive Clot	"I've ne...	Fibroids	"Massive...

```
# write_csv(posts_df, "reddit_fibroids_raw.csv")
```

```
# Combine title + text
```

```
posts_df$full_text <- paste(posts_df$title, posts_df$text)
```

```
posts_df$full_text <- iconv(posts_df$full_text,
  from = "latin1",
  to = "UTF-8",
  sub = "")
```

```
# Remove URLs
```

```
posts_df$full_text <- gsub("http[s]?://[^\ ]+", "", posts_df$full_text)
```

```
# Create quanteda corpus
```

```
reddit_corpus <- corpus(posts_df$full_text)
reddit_summary <- summary(reddit_corpus)
reddit_summary
```

Corpus consisting of 25 documents, showing 25 documents:

	Text	Types	Tokens	Sentences
text1		11	11	2
text2		5	6	3
text3		200	367	24
text4		147	241	15
text5		91	139	7
text6		47	67	5
text7		71	104	8
text8		23	26	1
text9		10	10	1
text10		108	159	7
text11		59	87	5
text12		279	670	24
text13		163	290	9
text14		105	164	8
text15		95	138	10
text16		74	100	4
text17		137	236	16
text18		175	299	15
text19		122	217	11
text20		17	17	2
text21		36	42	2
text22		59	72	5
text23		17	19	1
text24		79	124	10
text25		119	170	6

I converted text encoding to UTF-8, removed URLs, dropped punctuation and numbers, and filtered out common English stopwords. These steps preserved the actual meaning of what the women wrote while removing details that provide little analytical value.

After cleaning, I transformed the narratives into a quanteda corpus. The corpus summary reflected how many posts were available for analysis, providing insight into document length and structure. Each post became an analyzable unit, similar to chapters in a book, allowing me to track variation across individual experiences.

The corpus summary showed large variation in document size. Some posts contained only a few sentences, while others exceeded 600 tokens. This reflects the diversity of patient experiences: some women briefly share a symptom, while others describe months of complex recovery. This step created the structured foundation for subsequent linguistic analysis.

Tokenization, Stemming & Document-Feature Matrix (DFM)

At this stage, my goal was to convert the cleaned narratives into a structured format that allows for analysis. To do this, I tokenized the text, which involves breaking each post into individual words.

```
reddit_tokens <- tokens(
  reddit_corpus,
  remove_punct = TRUE,
  remove_numbers = TRUE
) %>%
tokens_tolower() %>%
tokens_select(
  pattern = c(
    stopwords("en"),
    "have","the","â","itâ","iâ","just","like","get","got","go",
    "hi","donâ","didnâ","s","t","m","ve"
  ),
  selection = "remove"
) %>%
tokens_wordstem(language = "en")

reddit_tokens
```

Tokens consisting of 25 documents.

text1 :

```
[1] "need"      "help"      "serious" "pls"      "see"      "red"      "mark"
```

text2 :

```
[1] "slynd"      "txa"      "aeygestin" "na"
```

text3 :

```
[1] "pain"      "manag"      "post"      "op"      "final"
[6] "found"      "doctor"      "nyu"      "like"      "go"
[11] "hybrid"      "myomectomi"
[ ... and 154 more ]
```

text4 :

```
[1] "need"      "hope"      "everyon" "post"      "time"      "long"      "journey"
[8] "found"      "fibroid" "connect" "blood"      "vessel"
[ ... and 82 more ]
```

text5 :

```
[1] "pregnanc" "stori"      "experi"      "larg"      "fibroid" "ladi"
[7] "cm"      "subseros" "fibroid"      "awar"      "year"      "husband"
[ ... and 42 more ]
```

text6 :

```
[1] "massiv" "clot"      "never"      "pass"      "clot"      "okay"      "realli" "okay"
[9] "sit"      "toilet" "liter"      "chang"
[ ... and 10 more ]
```

```
[ reached max_ndoc ... 19 more documents ]
```

```
# Create DFM
reddit_dfm <- dfm(reddit_tokens)
reddit_dfm
```

Document-feature matrix of: 25 documents, 703 features (93.16% sparse) and 0 docvars.

	features									
docs	need	help	serious	pls	see	red	mark	slynd	txa	aeystestin
text1	1	1	1	1	1	1	1	0	0	0
text2	0	0	0	0	0	0	0	1	1	1
text3	3	0	0	0	1	0	0	0	0	0
text4	1	0	0	0	0	0	0	0	0	0
text5	0	0	0	0	0	0	0	0	0	0
text6	0	0	0	0	0	0	0	0	0	0

[reached max_ndoc ... 19 more documents, reached max_nfeat ... 693 more features]

During tokenization, punctuation, numbers, and common stopwords were removed so that the analysis would focus on meaningful content rather than grammatical fillers.

In addition to standard cleaning steps, I applied word stemming. Stemming reduces words to their root form so that different grammatical variations of the same idea are treated as one. For example, words such as painful, pain, and pains are reduced to pain. The intention behind stemming was to ensure that conceptually identical experiences were counted together rather than split across multiple word forms. This is especially important in patient-authored narratives, where individuals may describe the same experience using different grammatical styles.

After tokenization and stemming, I converted the cleaned tokens into a document-feature matrix (DFM). In this matrix, each row(docs) represents one Reddit post, and each column(features) represents a unique word stem. The values in the matrix indicate how many times a given word stem appears in each post.

The DFM contained 25 documents and 703 unique word features.

Word Frequency Analysis

Word-frequency analysis helps identify what topics naturally dominate patient conversations without imposing pre-defined categories.

```
# WORD FREQUENCY
word_freq <- topfeatures(reddit_dfm, 20)
word_freq
```

fibroid	pain	remov	surgeri	month	also	period
41	17	17	15	13	12	12
one	day	need	doctor	year	uterus	tri
12	12	11	11	11	11	11
myomectomi	want	feel	back	time	anyon	
10	10	9	9	9	9	

7/27

around the condition itself, with pain emerging as one of several recurring concerns rather than the sole focus.

Together, these frequency patterns highlight how patient narratives intertwine physical symptoms, medical decision-making, and the passage of time during recovery.

Dictionary-Based Sentiment Analysis

Standard sentiment lexicons misclassify medical words (bleeding, cramp, etc) as emotional signals. To avoid this, I will create a custom sentiment dictionary that maps only emotionally meaningful words.

This step assesses the emotional tone embedded in recovery narratives.

```
# SENTIMENT DICTIONARY
my_sentiment_dict <- dictionary(list(
  positive = c("relief","improv","heal","better","hope","grate","strong", "pregnant", "re
  negative = c("pain","hurt","bleed","cramp","stress",
               "discourag","depress","wors","scare","fear", "bloat")
))

reddit_sent_dfm <- dfm_lookup(reddit_dfm, dictionary = my_sentiment_dict)

reddit_sent_df <- convert(reddit_sent_dfm, to = "data.frame")

reddit_scores <- reddit_sent_df %>%
  mutate(
    total = positive + negative,
    sentiment_score = ifelse(
      total == 0,
      0.5,
      positive / total
    )
  ) %>%
  select(doc_id, sentiment_score)

reddit_scores
```

	doc_id	sentiment_score
1	text1	0.5000000
2	text2	0.5000000
3	text3	0.0000000
4	text4	0.4000000
5	text5	1.0000000
6	text6	0.5000000
7	text7	0.5000000
8	text8	0.5000000
9	text9	0.5000000
10	text10	1.0000000
11	text11	0.0000000

12 text12	0.0000000
13 text13	0.0000000
14 text14	0.5000000
15 text15	0.0000000
16 text16	0.5000000
17 text17	0.0000000
18 text18	0.5555556
19 text19	0.7500000
20 text20	0.5000000
21 text21	0.0000000
22 text22	1.0000000
23 text23	0.5000000
24 text24	0.5000000
25 text25	0.5000000

To examine the emotional tone of the recovery narratives, I applied a custom sentiment dictionary tailored specifically to the medical context of myomectomy recovery. Rather than relying on standard sentiment lexicons which often misclassify medical terms, I restricted the dictionary to words that clearly signal emotional states. This approach ensured that sentiment scores reflected emotional expression rather than clinical description.

Using this dictionary, I counted the number of positive and negative emotional terms in each post and calculated a scaled sentiment score between 0 and 1. A score closer to 0 indicates predominantly negative emotion, 1 indicates predominantly positive emotion, and 0.5 represents a neutral or emotionally mixed post.

Overall Emotional Patterns

The sentiment scores reveal substantial variation across patient narratives, rather than a single dominant emotional tone.

Neutral posts (score = 0.5) make up a large portion of the dataset. These posts either contained no words from the sentiment dictionary or balanced positive and negative emotional terms. This suggests that many women focus on describing symptoms, timelines, or medical details without explicitly framing their experiences in emotional language.

Negative sentiment posts (scores near 0) appear in several narratives. These posts are dominated by words such as pain, fear, worse, and stress, indicating ongoing discomfort, anxiety, or frustration during recovery. The presence of multiple posts with scores of 0 shows that emotional distress remains a meaningful part of the recovery experience for some women.

Positive sentiment posts (scores near 1) are less frequent but clearly present. These narratives include words like hope, better, recover, and pregnant, reflecting relief, improvement, or optimism about future outcomes. Although fewer in number, these posts demonstrate that recovery is not universally negative and that some women experience emotional reassurance or progress.

Mixed sentiment posts (scores between 0.4 and 0.75) reflect varied experiences. For example, posts with scores such as 0.40, 0.56, and 0.75 indicate that women often express both difficulty and hope within the same narrative describing pain or uncertainty alongside improvement or emotional strength.

These results suggest that emotional recovery after myomectomy is not uniform. Many women share experiences that are emotionally neutral in tone, focused on monitoring symptoms or recovery timelines. Others express clear distress, while a smaller group emphasizes positive outcomes or emotional resilience.

Importantly, the dominance of neutral and negative scores does not imply the absence of hope. Instead, it reflects the reality that recovery is often framed in practical and clinical terms, with emotional language appearing most strongly when pain persists or when improvement becomes meaningful enough to note.

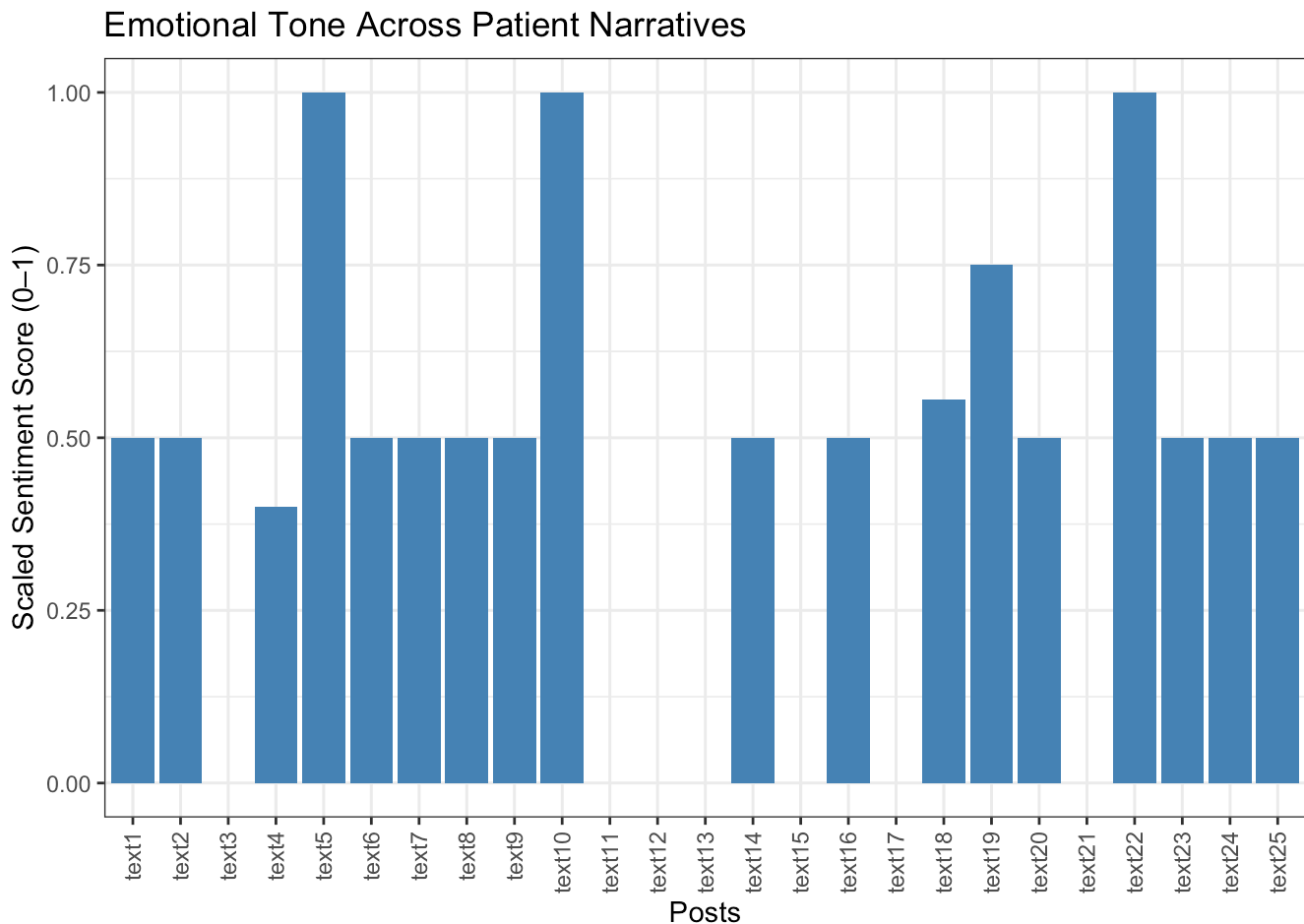
Overall, the sentiment analysis supports the idea that myomectomy recovery involves emotional variability, shaped by symptom persistence, uncertainty, and gradual improvement rather than a single emotional trajectory.

Visualizing Emotional Tone Across Narratives

To better understand the distribution of emotional expression across the dataset, I created a bar plot displaying the scaled sentiment score for each post. This visualization allows for a comparative view of how individual narratives differ in emotional tone.

```
reddit_scores$doc_id <- factor(
  reddit_scores$doc_id,
  levels = paste0("text", 1:nrow(reddit_scores))
)

# Bar plot of sentiment scores
ggplot(reddit_scores, aes(x = doc_id, y = sentiment_score)) +
  geom_col(fill = "steelblue") +
  theme_bw() +
  labs(
    title = "Emotional Tone Across Patient Narratives",
    x = "Posts",
    y = "Scaled Sentiment Score (0-1)"
  ) +
  theme(
    axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1)
  )
```



The bar plot reveals the degree of emotional expression across the 25 patient narratives. Several patterns are immediately visible:

Clustering of neutral scores (0.5): Approximately half of the posts show neutral sentiment, indicating that many women frame their experiences in descriptive rather than explicitly emotional language.

Presence of fully negative posts (0.0): Six posts contain exclusively negative emotional terms, suggesting significant distress or ongoing symptoms for a subset of patients.

Isolated positive peaks (1.0): Three posts scored 1.0, representing narratives dominated by hope, relief, or successful outcomes—though these are clearly in the minority.

Mixed sentiment posts: Posts with scores between 0.4 and 0.75 demonstrate that many women experience simultaneous hardship and hope, reflecting the complex emotional landscape of recovery. This visualization underscores that patient experiences cannot be reduced to a single emotional trajectory. Instead, recovery involves navigating uncertainty, managing persistent symptoms, and occasionally celebrating meaningful progress.

Topic Modeling

Word frequency analysis reveals which terms appear most often, but it cannot show how these terms cluster together to form coherent conversations. For instance, knowing that “pain” and “doctor” are both frequent does not tell us whether patients discuss pain management with doctors or mention these concepts in entirely separate contexts. To understand the underlying structure of patient discussions, what women actually talk about when they share their experiences, I employed topic

modeling. This approach identifies groups of words that consistently appear together across posts, revealing the distinct themes that shape recovery narratives in ways that simple word counts cannot capture.

```
library(stm)
```

stm v1.3.7 successfully loaded. See ?stm for help.

Papers, resources, and other materials at structuraltopicmodel.com

```
# Convert DFM to STM format

stm_input <- convert(reddit_dfm, to = "stm")

# Fit correlated topic model

set.seed(123)

cor_topic_model <- stm(

  documents = stm_input$documents,

  vocab = stm_input$vocab,

  K = 5,

  init.type = "Spectral",

  verbose = FALSE

)

# Inspect topics

summary(cor_topic_model)
```

A topic model with 5 topics, 25 documents and a 703 word dictionary.

Topic 1 Top Words:

Highest Prob: pain, manag, work, doc, need, year, said

FREX: doc, manag, success, work, pain, much, doctor

Lift: doc, 43f, 7-8, advoc, aeygestin, allerg, background

Score: aeygestin, doc, advoc, camera, like, opioid, put

Topic 2 Top Words:

Highest Prob: us, ultrasound, transvagin, call, gyn, routin, also

FREX: gyn, routin, us, transvagin, call, ultrasound, isnâ

Lift: 28th, anxieti, assess, blew, catch, check-up, click

Score: slynd, gyn, routin, ultrasound, us, hip, imag

Topic 3 Top Words:

Highest Prob: fibroid, remov, tri, ttc, uterus, surgeri, day

FREX: ttc, pregnanc, caviti, size, prior, tri, remov
 Lift: -, abil, abnorm, absolut, actual, adult, affect
 Score: remov, mark, ttc, tri, pregnanc, caviti, size

Topic 4 Top Words:

Highest Prob: fibroid, can, pain, also, pill, na, postpartum
 FREX: can, na, postpartum, pill, may, endometriosis, job
 Lift: endometriosis, job, therefor, +, acceler, action, age
 Score: txa, postpartum, endometriosis, job, therefor, age, pill

Topic 5 Top Words:

Highest Prob: fibroid, month, myomectomi, anyon, cramp, back, doctor
 FREX: cramp, month, els, inject, anyon, similar, extrem
 Lift: £, 10cm, 1cm, 2cm, 31f, 3cm, 4th
 Score: 2cm, month, els, inject, cramp, locat, anyon

```
labelTopics(cor_topic_model)
```

Topic 1 Top Words:

Highest Prob: pain, manag, work, doc, need, year, said
 FREX: doc, manag, success, work, pain, much, doctor
 Lift: doc, 43f, 7-8, advoc, aeygestin, allerg, background
 Score: aeygestin, doc, advoc, camera, like, opioid, put

Topic 2 Top Words:

Highest Prob: us, ultrasound, transvagin, call, gyn, routin, also
 FREX: gyn, routin, us, transvagin, call, ultrasound, isnâ
 Lift: 28th, anxieti, assess, blew, catch, check-up, click
 Score: slynd, gyn, routin, ultrasound, us, hip, imag

Topic 3 Top Words:

Highest Prob: fibroid, remov, tri, ttc, uterus, surgeri, day
 FREX: ttc, pregnanc, caviti, size, prior, tri, remov
 Lift: -, abil, abnorm, absolut, actual, adult, affect
 Score: remov, mark, ttc, tri, pregnanc, caviti, size

Topic 4 Top Words:

Highest Prob: fibroid, can, pain, also, pill, na, postpartum
 FREX: can, na, postpartum, pill, may, endometriosis, job
 Lift: endometriosis, job, therefor, +, acceler, action, age
 Score: txa, postpartum, endometriosis, job, therefor, age, pill

Topic 5 Top Words:

Highest Prob: fibroid, month, myomectomi, anyon, cramp, back, doctor
 FREX: cramp, month, els, inject, anyon, similar, extrem
 Lift: £, 10cm, 1cm, 2cm, 31f, 3cm, 4th
 Score: 2cm, month, els, inject, cramp, locat, anyon

```
# Representative documents
```

```
findThoughts(  
  
cor_topic_model,  
  
texts = posts_df$full_text,
```

```
topics = 1:5,  
  
n = 1  
  
)
```

Topic 1:

Pain management post op Hi! I finally found a doctor I like at NYU for what is likely going to be a hybrid myomectomy. Bikini cut, no cameras, experienced doc. She doesn't use cameras and feels confident about success.

43f, 8 fibroids, intramural, largest are like 7-8 cm.

I stopped seeing the last doc at Columbia because she was a Tylenol only doc. No thank you. Present doc said she will manage my pain well and really likes gabapentin. My 11 year old daughter had a major surgery last year and said it barely touched her pain. I had to fight doctors to get her pump back after she screamed bloody murder for hours because she had bolts put in her spine and they just didn't think it was that bad. I'm tired of this.

Doc said the nerve blocks don't really work. She offered another older NSAID and I'm getting worried.

I'm a redhead who is allergic to Motrin and need advice on how you all successfully advocated for proper pain management.

I've went without dental before for years long stretches because of a severe lack of proper pain management. My epidural to vaginally deliver my 11.3 lb kid only worked on half my body. I have too much trauma around pain and have put the surgery off because I couldn't find a doctor who wasn't on the Tylenol bus. It takes a lot to numb me, a lot to knock me out, and opioids worked well the three times in my entire life I've needed them. No issues. I also hold a city gov position and my background is sterling.

I need advice on what worked and how you managed to successfully advocate.

Unhelpful responses include but are not limited to: it wasn't that bad and I didn't take opioids and it worked fine. That is not me.

I won't do it unless I have access. Thanks so much!

Topic 2:

Shocked to learn that transvaginal ultrasounds aren't routine in US gyn visits I was listening to a podcast episode the other day and there was a gynecologist on. At one point she said that in the US a transvaginal ultrasound isn't part of a routine gyn visit and I was honestly shocked.

I live in Italy and here it's totally standard like, you go in for your regular check-up and you get a transvaginal ultrasound just like you get a Pap smear. So hearing that this isn't normal in the US kind of blew my mind.

It also suddenly explains why there seem to be so many cases of fibroids that go undiagnosed until they're massive. If they're not doing ultrasounds regularly, how would they catch them early?

Is this really the norm in the US? I'm genuinely curious how routine gyn care works over there

Topic 3:

Fibroids on the outside of uterus while ttc I went to my gyno for my annual transvaginal ultra sound and my well woman exam and I wanted to ask her if it was safe to start ttc. I'm 32 years old and I know last year I had a few fibroids up to 5 cm but no one knew their exact placement since I hadn't had an MRI yet.

My doctor is apparently the best at this type of surgery so I trust her but I just always am a bit confused as to what to do because last year I wasn't married and wasn't ttc yet so she didn't really go into detail about removal and also at 5 cm they aren't considered that big.

During my visit she said it seems like one of them is about 6cm. We usually remove them around 7 cm. Yours seem to be hanging off of the uterus on the outside but we should do an mri to confirm this. If the fibroids are not distorting the cavity you should be fine to try. We try to not perform surgery unless absolutely necessary because then you will need a C-section and that's 2 surgeries. You should try to have the kids you want and then opt for removal and possibly remove the whole uterus at that point. Given how young you are and their size, surgery is going to be needed at some point in your life

So I went and got an MRI and the doctor found that actually I have 2 fibroids that are 7 cm and one fibroid that is small. All are pedunculated on the outside of my uterus but one is growing out from the mid muscle. She said, the good news is that none of your fibroids are growing inside the cavity. Your cavity is clear and not being distorted. The bad news is that this is around the size where we consider removal prior to pregnancy, but since you want to ttc right away, I think it should be fine to try given that you know the pros and cons to each option

I replied to her message with a few questions.

Why would someone remove fibroids that are on the outside of the uterus prior to pregnancy if they are not distorting the cavity?

If they continue to grow during my ttc journey do I then need to have them removed since they are at the size you normally remove them?

If I want 2 kids should I remove them after the first kid before trying for a 2nd? – she told me in the office that I should remove them when I'm done having kids but then after the MRI her mentioning she normally removed them at my size made me feel conflicted about that.

Is removal of this type of fibroid less invasive than fibroids on the inside of the cavity?

I'm still waiting to hear back with her replies to my questions but I wanted to come on here and ask if anyone has experience with ttc while fibroids are large but only on

the outside of the uterus. I've read that these can be painful for the mother but don't affect fertility. And my doctor kind of nodded her head yes when I told her that. So then why would you even need to remove those prior to pregnancy? I am a woman who goes through physical pain all the time so I'm not worried about pain my main concern is my ability to have children.

Edit:

Thank you so so SO much for your replies and sharing your personal experiences. It has truly helped me a lot.

Topic 4:

Desogestrel I got diagnosed with 3 fibroids in 2024 after a couple of visits to A&E. After MRI and CT scan + ultrasounds it was concluded that I have 2 intramural and 1 pedunculated fibroids, one of them was degenerating and probably causing the excruciating pain, gyno also considers I may have endometriosis, the MRI didn't show clear lesions and I didn't want an exploratory surgery at that stage, I therefore don't have a diagnosis for endometriosis. So after considering my options I went to the pill and it has been longer than a year on it. Now I have been experiencing pain on my left arm which I know can be a symptom of a clot so I asked for an appointment with my gp to discuss course of action.

I have a dilemma here! I enjoy not having my period and feel more in control of my emotions to be honest. I gained a little bit of weight but still healthy weight and obviously no pain with my period. I'm terrified of stopping the pill and having pain again that can make me lose days of work and therefore my job because this job I have now it's very strict with sick days and I'm also under 2 years so they can sack me for any reason (uk). I don't know what to do but I also don't want a hysterectomy or hormones to accelerate menopause. It seems like stopping the pill it's the safest thing to do but this may cost me my job!

What would you do?

Topic 5:

Need some hope Hi Everyone,

I've posted on here a few times but this has been a long journey. They found a fibroid that is connected to blood vessels from my uterus and it's sitting just above my cervix on the left side.

I have been in terrible pain anytime I have any sexual activity that results in an orgasm. On top of it, I get severe cramping when overexerting myself or lifting anything. I'm also getting my period twice a month now with terrible cramping and blood clots/loss.

Due to my weight and the location of the fibroid they are not comfortable doing a minimally invasive myomectomy. The only answer I've really gotten is a hysterectomy.

They referred me to a second opinion that took all my insurance info and booked the appointment 2 months ago for this upcoming Monday. They called me today and said they don't take my insurance and they are cancelling the appointment. At this point I have lost all hope I had. I'm 34, no children, and am not ready to go the hysterectomy route. It feels like I have now been dumped to the wayside for 2 months and nobody is listening or understanding me.

Any advice? Anyone been in a similar boat? Iâ€™ m struggling hard ð

```
# Create covariate

posts_df$ttc <- as.integer(grepl("ttc", posts_df$full_text))

posts_df$ttc <- factor(posts_df$ttc)

# Fit STM with prevalence

set.seed(123)

myModel <- stm(

  documents = stm_input$documents,

  vocab = stm_input$vocab,

  K = 5,

  prevalence = ~ ttc,

  data = posts_df,

  max.em.its = 1000,

  init.type = "Spectral"

)
```

Beginning Spectral Initialization

Calculating the gram matrix...

Finding anchor words...

.....

Recovering initialization...

.....

Initialization complete.

.....

Completed E-Step (0 seconds).

Completed M-Step.

Completing Iteration 1 (approx. per word bound = -6.133)

.....

Completed E-Step (0 seconds).

Completed M-Step.

Completing Iteration 2 (approx. per word bound = -6.027, relative change = 1.740e-02)

.....

Completed E-Step (0 seconds).

Completed M-Step.

Completing Iteration 3 (approx. per word bound = -5.963, relative change = 1.059e-02)

```
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 4 (approx. per word bound = -5.906, relative change = 9.461e-03)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 5 (approx. per word bound = -5.870, relative change = 6.191e-03)
Topic 1: work, doctor, doc, need, advic
Topic 2: work, doctor, doc, need, advic
Topic 3: fibroid, remov, pain, period, experi
Topic 4: can, need, mri, consid, right
Topic 5: surgeri, month, one, tri, anyon
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 6 (approx. per word bound = -5.835, relative change = 5.931e-03)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 7 (approx. per word bound = -5.801, relative change = 5.735e-03)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 8 (approx. per word bound = -5.728, relative change = 1.258e-02)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 9 (approx. per word bound = -5.663, relative change = 1.142e-02)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 10 (approx. per word bound = -5.627, relative change = 6.329e-03)
Topic 1: us, call, transvagin, gyn, routin
Topic 2: manag, work, doc, success, need
Topic 3: fibroid, remov, week, ttc, cm
Topic 4: can, need, mri, consid, right
Topic 5: fibroid, pain, surgeri, month, one
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 11 (approx. per word bound = -5.585, relative change = 7.524e-03)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 12 (approx. per word bound = -5.536, relative change = 8.663e-03)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 13 (approx. per word bound = -5.501, relative change = 6.396e-03)
.....
```

Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 14 (approx. per word bound = -5.468, relative change = 6.035e-03)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 15 (approx. per word bound = -5.440, relative change = 5.172e-03)
Topic 1: us, ultrasound, call, transvagin, gyn
Topic 2: pain, manag, work, doc, year
Topic 3: fibroid, remov, tri, ttc, uterus
Topic 4: pain, can, consid, also, mri
Topic 5: fibroid, month, surgeri, anyon, one
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 16 (approx. per word bound = -5.416, relative change = 4.290e-03)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 17 (approx. per word bound = -5.402, relative change = 2.582e-03)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 18 (approx. per word bound = -5.394, relative change = 1.616e-03)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 19 (approx. per word bound = -5.384, relative change = 1.859e-03)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 20 (approx. per word bound = -5.374, relative change = 1.718e-03)
Topic 1: us, ultrasound, call, transvagin, gyn
Topic 2: pain, manag, work, doc, year
Topic 3: fibroid, remov, tri, ttc, uterus
Topic 4: surgeri, pain, fibroid, can, also
Topic 5: fibroid, month, anyon, myomectomi, bleed
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 21 (approx. per word bound = -5.366, relative change = 1.457e-03)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 22 (approx. per word bound = -5.360, relative change = 1.191e-03)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 23 (approx. per word bound = -5.355, relative change = 8.775e-04)
.....
Completed E-Step (0 seconds).

```

Completed M-Step.
Completing Iteration 24 (approx. per word bound = -5.353, relative change = 5.148e-04)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 25 (approx. per word bound = -5.350, relative change = 5.177e-04)
Topic 1: us, ultrasound, transvagin, call, gyn
Topic 2: pain, manag, work, doc, year
Topic 3: fibroid, remov, tri, ttc, uterus
Topic 4: fibroid, pain, can, also, pill
Topic 5: fibroid, month, myomectomi, anyon, cramp
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 26 (approx. per word bound = -5.347, relative change = 4.661e-04)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 27 (approx. per word bound = -5.345, relative change = 4.123e-04)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 28 (approx. per word bound = -5.342, relative change = 5.173e-04)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 29 (approx. per word bound = -5.341, relative change = 2.254e-04)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 30 (approx. per word bound = -5.341, relative change = 4.163e-05)
Topic 1: us, ultrasound, transvagin, call, gyn
Topic 2: pain, manag, work, doc, need
Topic 3: fibroid, remov, tri, ttc, uterus
Topic 4: fibroid, can, pain, also, pill
Topic 5: fibroid, month, myomectomi, anyon, cramp
.....
Completed E-Step (0 seconds).
Completed M-Step.
Completing Iteration 31 (approx. per word bound = -5.341, relative change = 1.529e-05)
.....
Completed E-Step (0 seconds).
Completed M-Step.
Model Converged

```

```
# Topic inspection
```

```
labelTopics(myModel)
```

Topic 1 Top Words:

Highest Prob: us, ultrasound, transvagin, call, gyn, routin, also

FREX: gyn, routin, us, transvagin, call, ultrasound, isnâ

Lift: 28th, aeygestin, anxieti, assess, blew, catch, check-up

Score: aeygestin, gyn, routin, ultrasound, us, hip, imag

Topic 2 Top Words:

Highest Prob: pain, manag, work, doc, need, year, said

FREX: doc, manag, success, work, pain, much, doctor

Lift: 43f, 7-8, advoc, allerg, background, bare, bikini

Score: slynd, doc, advoc, camera, like, opioid, put

Topic 3 Top Words:

Highest Prob: fibroid, remov, tri, ttc, uterus, surgeri, day

FREX: ttc, pregnanc, caviti, prior, size, tri, remov

Lift: -, abil, abnorm, absolut, actual, adult, affect

Score: remov, mark, ttc, tri, pregnanc, caviti, size

Topic 4 Top Words:

Highest Prob: fibroid, can, pain, also, pill, na, postpartum

FREX: can, na, postpartum, pill, may, endometriosi, job

Lift: endometriosi, job, therefor, +, acceler, action, age

Score: txa, postpartum, endometriosi, job, therefor, age, pill

Topic 5 Top Words:

Highest Prob: fibroid, month, myomectomi, anyon, cramp, back, doctor

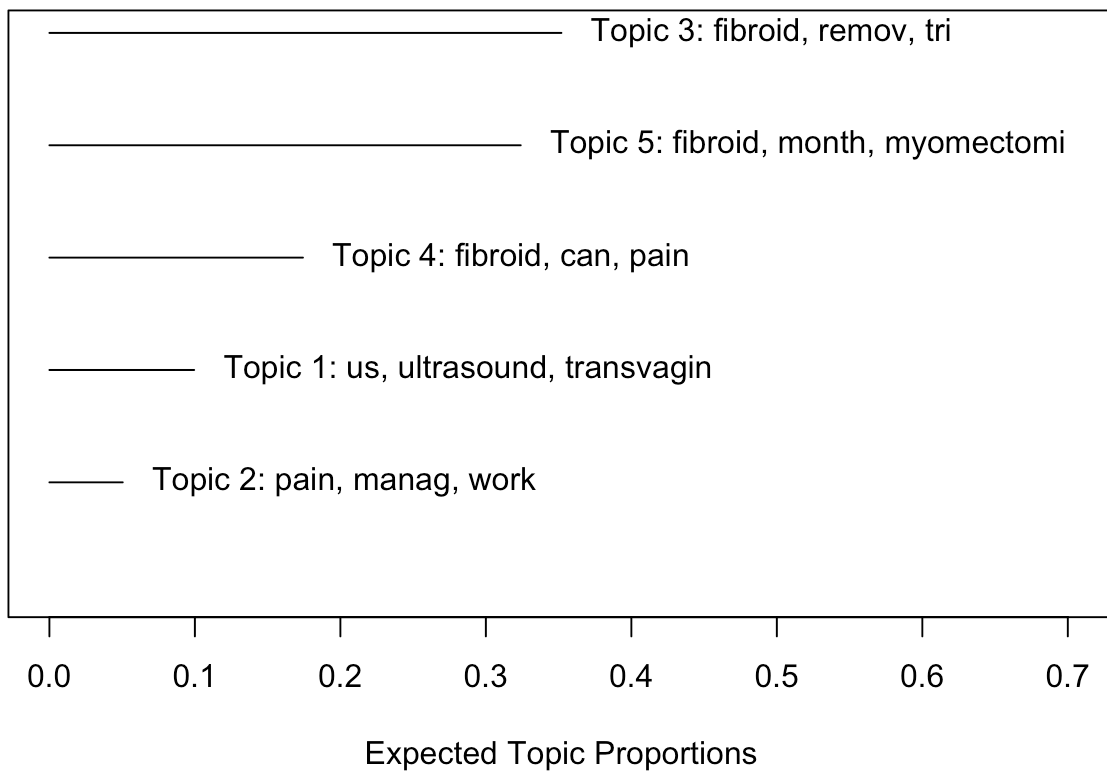
FREX: cramp, month, els, inject, anyon, similar, extrem

Lift: £, 10cm, 1cm, 2cm, 31f, 3cm, 4th

Score: 2cm, month, els, inject, cramp, locat, anyon

```
plot(myModel, type = "summary")
```

Top Topics



```
k <- 5

myTopicNames <- labelTopics(myModel, n = 4)$frex

myTopicLabels <- rep(NA, k)

for (i in 1:k) {

  myTopicLabels[i] <- paste(myTopicNames[i, ], collapse = "_")

}

myTopicLabels
```

```
[1] "gyn_routin_us_transvagin" "doc_manag_success_work"
[3] "ttc_pregnanc_caviti_prior" "can_na_postpartum_pill"
[5] "cramp_month_els_inject"
```

```
modelEffects <- estimateEffect(

  formula = 1:k ~ ttc,

  stmobj = myModel,
```

```
metadata = posts_df

)

par(mfrow = c(2, 3), bty = "n", lwd = 2)

for (i in 1:k) {

plot.estimateEffect(

modelEffects,

covariate = "ttc",

topics = i,

method = "difference",

cov.value1 = 1,

cov.value2 = 0,

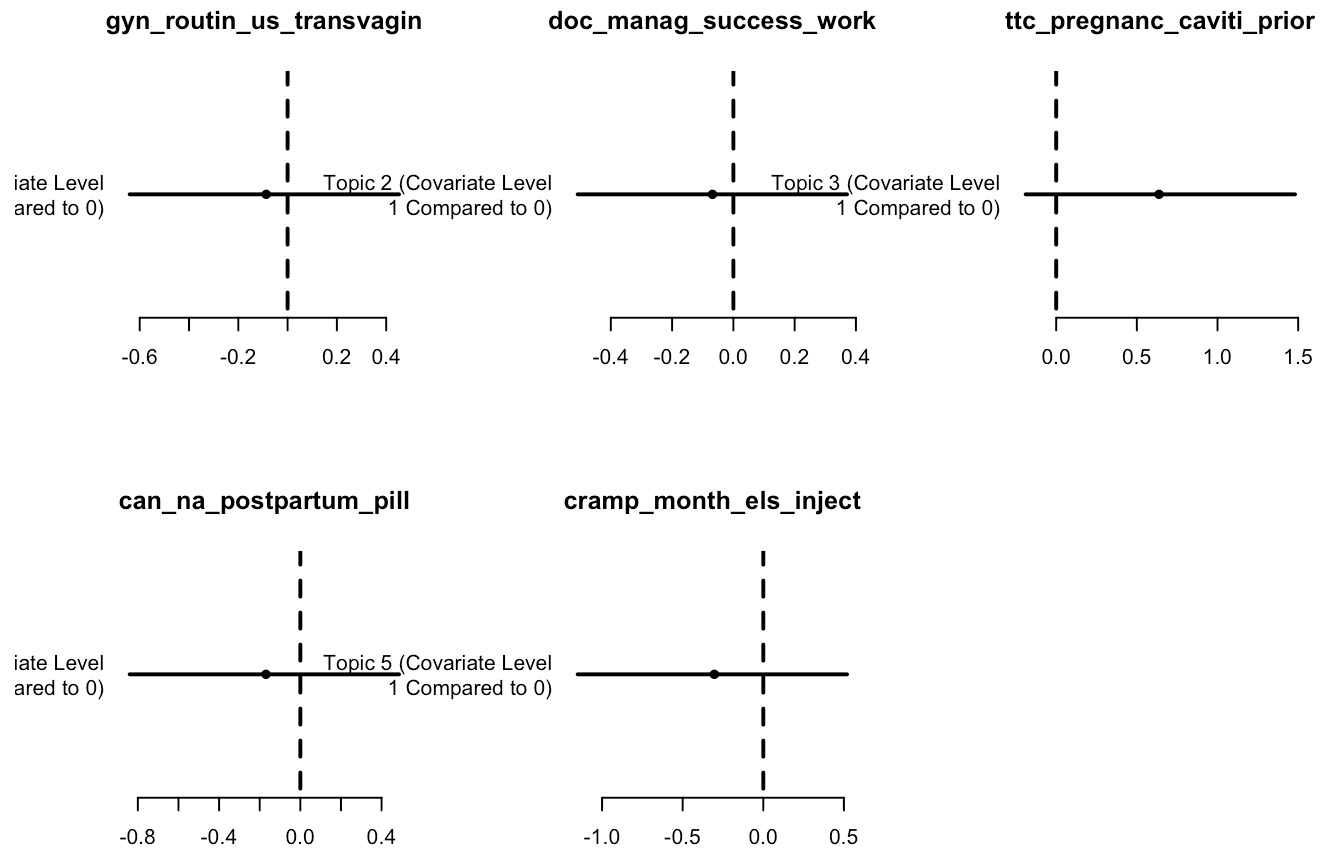
model = myModel,

main = myTopicLabels[i],

printlegend = FALSE

)

}
```



The structural topic model identifies five substantively distinct themes in the posts, each corresponding to a recurring aspect of patients' experiences and concerns.

Topic 3, characterized by the terms *fibroid*, *remov*, and *tri*, represents discussion of fibroid removal procedures and surgical decision-making, often framed around trial attempts at treatment or uncertainty about next steps. This topic has the highest expected prevalence, indicating that surgical intervention is the dominant concern in the corpus.

Topic 5, defined by *fibroid*, *month*, and *myomectomy*, captures longer-term recovery timelines and post-procedure monitoring. Posts associated with this topic tend to focus on monthly cycles, symptom changes over time, and expectations following myomectomy, suggesting an emphasis on recovery rather than acute intervention.

Topic 4, marked by *fibroid*, *can*, and *pain*, reflects symptom-centered narratives. These posts focus on pain management, daily functioning, and the physical burden of fibroids, highlighting how symptoms shape quality of life and motivate treatment discussions.

Topic 1, associated with *us*, *ultrasound*, and *transvaginal*, centers on diagnostic processes. This topic captures conversations about imaging, test results, and clinical evaluations, indicating that many users are seeking clarity and confirmation through medical diagnostics.

Topic 2, defined by *pain*, *manag*, and *work*, represents the intersection of health and everyday responsibilities. Posts in this topic emphasize managing pain while maintaining work and routine

activities, underscoring the practical constraints patients face when coping with chronic symptoms.

The prevalence estimates show that surgical and recovery-related topics (Topics 3 and 5) dominate the discussion, while diagnostic and work-management topics are less prevalent but still substantively important. Together, these results indicate that the discourse is not simply about having fibroids, but about navigating diagnosis, enduring symptoms, making treatment decisions, and managing the long-term impact on daily life.

Conclusion

This text analysis of 25 Reddit posts from r/Fibroids provides a patient-centered perspective on myomectomy recovery that complements traditional clinical research.

Key Findings

Language Centers on the Condition Itself

The term “fibroid” appeared 41 times—more than twice as often as any other word—indicating that patients consistently frame their experiences around their diagnosis rather than any single symptom or outcome.

Pain Remains Prominent but Not Dominant

Pain-related terms appeared frequently (17 mentions), alongside surgical terminology (remov: 17, surgeri: 15) and time-related words (month: 13, day: 12, year: 11). This suggests that recovery narratives interweave physical discomfort with medical decision-making and temporal tracking of progress.

Emotional Tone is Variable, Not Uniformly Positive

Sentiment analysis revealed substantial heterogeneity. While some posts expressed clear positivity (scores of 1.0), many were neutral (0.5), and several reflected predominantly negative emotional language (scores of 0.0). This pattern indicates that emotional recovery after myomectomy is neither linear nor universally positive.

Five Distinct Themes Emerged

Topic modeling identified conversations centered on (a) diagnostic imaging and ultrasounds, (b) pain management while maintaining work responsibilities, (c) fertility concerns and trying to conceive, (d) hormonal treatments and symptom control, and (e) post-myomectomy recovery timelines and cramping. Surgical and recovery-related themes showed the highest prevalence.

Patients Seek Community and Validation

Words like “anyon” (anyone), “need,” and “want” appeared frequently, reflecting how women use online forums not only to share information but also to seek reassurance, advice, and confirmation that their experiences are normal.

Hypothesis Evaluation

The initial hypothesis predicted that most women would describe improvement after myomectomy, such as reduced pain or bleeding. This hypothesis was only partially supported. While some narratives did express relief, hope, and recovery, the overall sentiment distribution showed that negative and neutral emotional tones were more common than positive ones. Pain-related language remained prominent throughout the dataset, and topic modeling revealed that symptom persistence and management challenges were central themes.

These findings suggest that while myomectomy can lead to positive outcomes, the recovery experience as expressed in online narratives is often characterized by ongoing uncertainty, residual symptoms, and emotional strain rather than unambiguous improvement. This may reflect the self-selection bias inherent in online health communities, where individuals experiencing complications or seeking support may be more motivated to post than those with straightforward recoveries.

Limitations

Several methodological constraints should be considered when interpreting these findings:

Sample Size and Representativeness

The analysis was based on only 25 Reddit posts, which limits statistical power and generalizability. Reddit users represent a specific demographic—typically younger, more digitally literate, and English-speaking—and may not reflect the broader population of women undergoing myomectomy.

Self-Selection Bias

Individuals who post about medical experiences online may differ systematically from those who do not. Women experiencing complications, prolonged recovery, or emotional distress may be more motivated to seek community support, potentially inflating the prevalence of negative narratives in the dataset.

Stemming Artifacts

Word stemming occasionally produced ambiguous root forms (e.g., “tri” could derive from “try,” “trial,” or “trying”), potentially combining distinct concepts. Manual inspection was required to interpret some findings.

Cross-Sectional Design

The analysis captures a snapshot of posts at one point in time and cannot track how individual patients’ experiences or sentiments evolve over the course of recovery.

Suggestions for Improvement and Future Research

Expand the Dataset

Future studies should collect a larger corpus spanning multiple time periods to improve statistical reliability and capture temporal trends in patient discourse.

Incorporate Longitudinal Tracking

By identifying users who post multiple times, researchers could examine how sentiment and concerns evolve throughout the recovery journey.

Compare Across Platforms and Procedures

Extending the analysis to other platforms (e.g., HysterSisters, Facebook groups) and comparing myomectomy narratives to those describing hysterectomy or uterine fibroid embolization could reveal procedure-specific patterns.

References

De Choudhury, M., Morris, M., & White, R. (2014). Seeking and sharing health information online. *ACM Transactions on Computer-Human Interaction*, 21(1), 1–32.

Stewart, E. A. (2015). Uterine fibroids. *The New England Journal of Medicine*, 372, 1646–1655.